TEXT IMPLEMENTATION

NORMAL RESEARCH PAPER CHA APPROACH USE KELA

1. Implementation details

Goal: Restaurant reviews ko 3 classes mein classify karna (Negative / Neutral / Positive) using transfer learning.

Pipeline (5 phases):

1. Dataset load + EDA
2. Label mapping + balanced sampling + 80-20 split
3. BERT (bert-base-uncased) training
4. RoBERTa (roberta-base) training
5. Results table + visualizations

Model architecture (SentimentClassifier):

- Pretrained backbone (BERT/RoBERTa) via AutoModel
- [CLS] token ka hidden state (768-dim) lena
- Dropout (0.3)
- Linear layer (768 → 3 classes)

Do training modes:

- **Frozen:** backbone ke weights freeze, sirf classifier head train hota hai
- **Fine-tuned:** poora model end-to-end train hota hai

Hyperparameters:

Parameter	**Frozen**	**Fine-tuned**
Epochs	4	3
Learning rate	1e-3	2e-5
Optimizer	Adam	Adam
Loss	CrossEntropy	CrossEntropy
Batch size	16	16
Max sequence length	128	128

Parameter	Frozen	Fine-tuned
Seed	42	42

Environment: Kaggle, GPU (CUDA), PyTorch + HuggingFace Transformers, scikit-learn metrics.

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2. Dataset aur Data Preprocessing

Dataset: Malaysia Restaurant Review Dataset (choonkhonng, Kaggle). Isme Google Review aur TripAdvisor reviews 10 cities ke hain (KL, Penang, JB, Ipoh, Melaka, Kuching, Miri, Langkawi, Shah Alam, Petaling Jaya). Use hua: GoogleReview_data_cleaned.csv

Preprocessing steps:

1. Sirf Review aur Rating columns rakhe
2. Missing values drop kiye (dropna)
3. Whitespace strip, empty reviews hataye
4. Duplicate reviews drop kiye
5. **Rating → Sentiment mapping:** 4-5 = Positive, 3 = Neutral, 1-2 = Negative
6. **Class balancing:** har class se 5000 random samples (total 15,000), paper ke setup ke hisaab se
7. **Label encoding:** Negative=0, Neutral=1, Positive=2
8. **Stratified 80-20 split:** 12,000 train / 3,000 test (har class 4000 / 1000)
9. **Tokenization:** model ka apna tokenizer, truncation + padding to 128 tokens

Stage	Rows
Original	222,020
After cleaning	199,171
Balanced	15,000
Train / Test	12,000 / 3,000

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3. Exploration (EDA) methods

- df.shape, df.columns, df.head(), df.dtypes se structure check

- df.isnull().sum() se missing values
- Review ka **word count** (average aur max) nikalne ka code
- Rating ke unique values check
- **Class distribution** (value_counts) full data par aur balanced data par

Key finding: original data **heavily imbalanced** hai:

Class	Count
Positive	156,390
Neutral	22,376
Negative	20,405

Isi wajah se balanced sampling (undersampling) kiya gaya. Paper/blackbook mein ye zaroor likhna.

(Note: notebook ke saved output mein word-count stats aur missing-value counts nahi dikh rahe, wo EDA cell ka output truncate/overwrite ho gaya hai. Agar numbers chahiye toh us cell ko dobara chalake output bhej dena.)

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4. Results

Test set: 3,000 samples, weighted metrics.

Model	Mode	Accuracy	Precision	Recall	F1	Time (s)
BERT	Frozen	67.37	67.43	67.37	67.40	453.2
BERT	Fine-tuned	72.17	71.55	72.17	71.44	900.0
RoBERTa	Frozen	66.37	66.27	66.37	66.28	446.6
RoBERTa	Fine-tuned	**75.07**	**75.20**	**75.07**	**75.13**	906.1

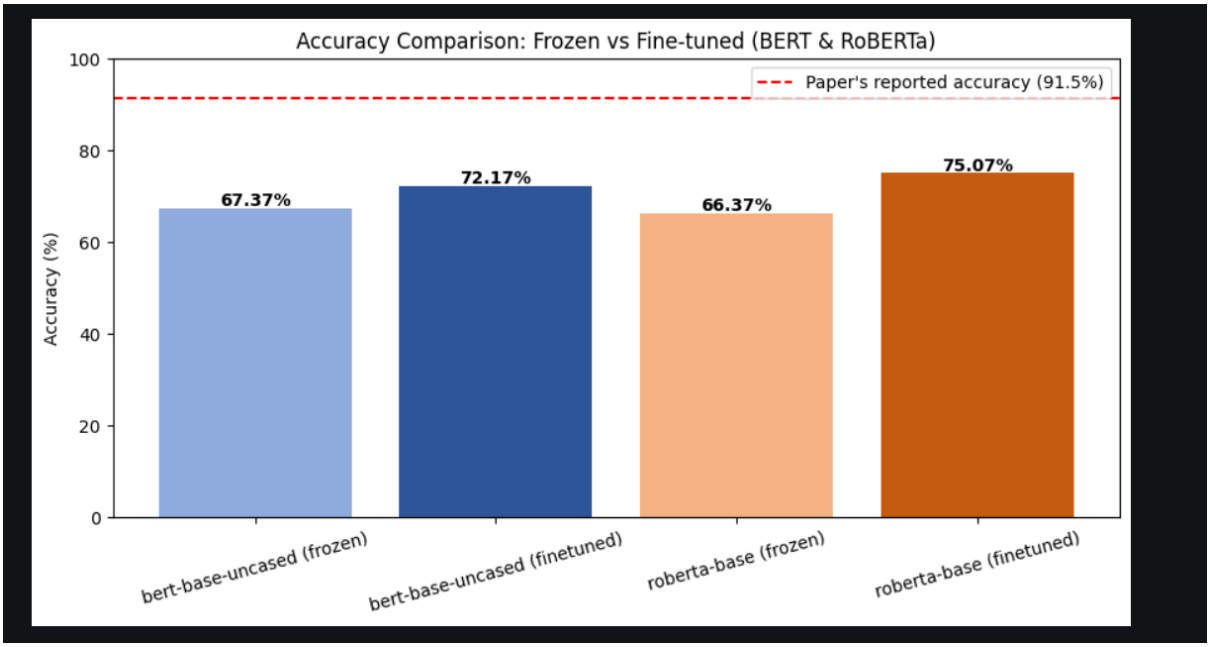
Training loss (epoch-wise):

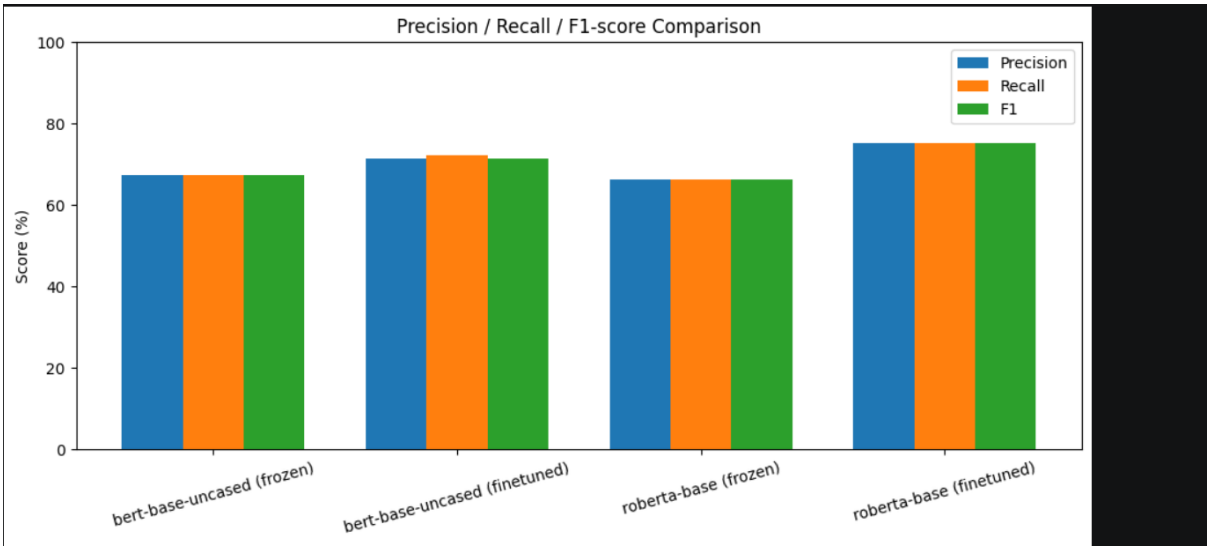
- BERT frozen: 0.858 → 0.802 → 0.800 → 0.792
- BERT fine-tuned: 0.695 → 0.529 → 0.377
- RoBERTa frozen: 1.052 → 0.990 → 0.958 → 0.931
- RoBERTa fine-tuned: 0.699 → 0.574 → 0.491

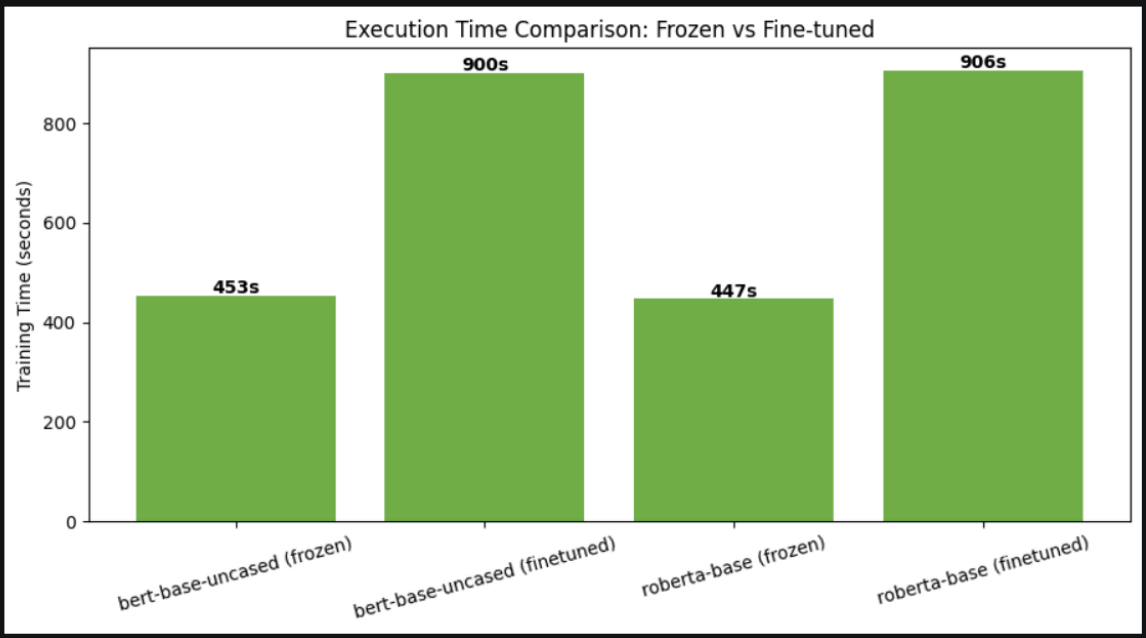
Fine-tuning ka improvement: BERT +4.8%, RoBERTa +8.7%.

Observations / discussion points:

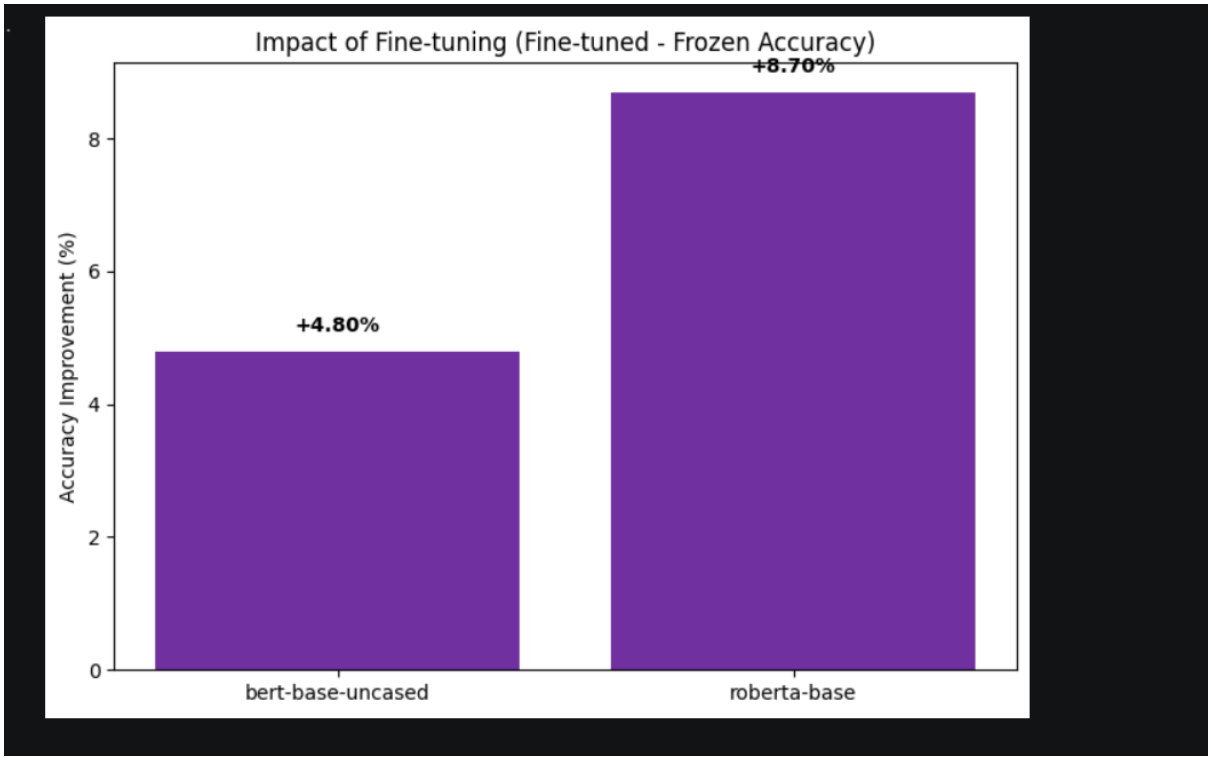
1. Fine-tuning dono models mein frozen se behtar hai, lekin training time lagbhag **2x** lagta hai.
2. **RoBERTa fine-tuned best** hai (75.07%), BERT fine-tuned se ~2.9% upar.
3. Frozen mode mein BERT thoda behtar hai RoBERTa se (67.37 vs 66.37). Reason: frozen RoBERTa ke representations [CLS] (<s>) token par classification ke liye utne ready nahi hote, aur uska pooler layer pretrained checkpoint mein nahi tha.
4. Neutral class generally sabse mushkil hoti hai kyunki 3-star reviews ka sentiment mixed hota hai (ye expected reason hai, notebook mein per-class analysis nahi hai).
5. **Paper se gap:** paper ka text unimodal accuracy **91.50%** hai, aapka best 75.07%. Honest reason likh sakte ho: alag dataset/labeling (paper sentiment score thresholds use karta hai, aapne star ratings), sirf 3-4 epochs, no hyperparameter tuning, aur 3-class star-rating-based labels noisy hote hain.








```
Accuracy improvement from fine-tuning:
 mode              finetuned  frozen  improvement
model
bert-base-uncased      72.17   67.37          4.8
roberta-base           75.07   66.37          8.7
```



NANTAR MAGH AAPLA APPROACH TUN IMPLEMENT KELA

1. Implementation details

Goal aur pipeline same hai: Malaysia restaurant reviews ka 3-class sentiment (Negative / Neutral / Positive) using BERT aur RoBERTa, frozen vs fine-tuned.

Architecture same hai: pretrained backbone → [CLS] hidden state → Dropout(0.3) → Linear(768→3).

v1 se v2 mein kya badla (ye blackbook/paper mein ek table banake zaroor daalna):

Parameter	**v1**	**v2 (updated)**	**Reason**
MAX_LEN	128	**256**	kam truncation, lambe reviews capture
Batch size	16	**32**	better GPU use, stable gradients
LR (frozen)	1e-3	**5e-4**	purana LR zyada tha, loss kam move kar raha tha
LR (fine-tuned)	2e-5	**3e-5**	standard fine-tuning range
Epochs (fine-tuned)	3	**4**	warmup ke liye thode zyada steps

Parameter	v1	v2 (updated)	Reason
Epochs (frozen)	4	4	same
Scheduler	none	**Linear warmup + decay** (10% warmup)	training stability
Gradient clipping	none	**max_norm = 1.0**	exploding gradients se bachav
Optimizer / Loss	Adam / CrossEntropy	same	
Epoch-wise tracking	loss only	**loss + test accuracy**	curves plot karne ke liye

Note: paper ke Table 3 ke hyperparameters (LR 0.001 BiLSTM, 0.00001 CNN) paper ke apne custom models ke hain, pretrained BERT/RoBERTa ke liye nahi. Isliye yahan standard fine-tuning practices use hui (ye bat notebook ke docstring mein bhi likhi hai, report mein bhi likh dena).

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2. Data preprocessing

Bilkul same hai v1 jaisa (Phase 1 aur 2 ka code aur output identical hai):

1. Sirf Review aur Rating columns, dropna
2. Whitespace strip, empty reviews hataye, duplicates drop
3. Rating → Sentiment: 4-5 Positive, 3 Neutral, 1-2 Negative
4. Balanced sampling: 5000 per class (15,000 total), seed 42
5. Label encoding: Negative=0, Neutral=1, Positive=2
6. Stratified 80-20 split: 12,000 train / 3,000 test
7. Tokenization with model tokenizer, truncation + padding to **256** tokens (v2 mein)

Stage	Rows
Original (Google Reviews cleaned)	222,020
After cleaning	199,171
Full distribution	Positive 156,390 / Neutral 22,376 / Negative 20,405

Stage	Rows
Balanced	15,000
Train / Test	12,000 / 3,000

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3. Exploration (EDA) methods

Same as before: shape, columns, dtypes, head, missing values, review word-count (average/max), unique ratings, aur class distribution (full vs balanced). Key finding same hai: dataset **heavily imbalanced** (Positive ~78%), isliye undersampling.

Phase 1 ka output is notebook mein sirf file listing tak saved hai, word-count aur missing-value stats nahi dikhte. Wo numbers chahiye toh Phase 1 cell dobara run karke output bhej dena.

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4. Results

BERT v2 (test set, 3,000 samples, weighted metrics)

Model	Mode	Accuracy	Precision	Recall	F1	Time (s)
BERT	Frozen	66.67	66.08	66.67	66.20	867.6
BERT	Fine-tuned	71.47	72.01	71.47	71.64	2114.1
RoBERTa	Frozen	pending				
RoBERTa	Fine-tuned	pending				

Epoch-wise (naya: loss + test accuracy)

Epoch	BERT frozen loss / acc	BERT fine-tuned loss / acc
1	0.9558 / 65.57%	0.7557 / **72.93%**
2	0.8220 / 66.33%	0.5360 / 72.77%
3	0.8056 / 66.37%	0.3866 / 71.67%
4	0.7933 / 66.67%	0.2576 / 71.47%

v1 vs v2 (BERT)

Config	v1 acc	v2 acc	Change
Frozen	67.37	66.67	-0.70
Fine-tuned	72.17	71.47	-0.70

Honest observations (inhe discussion mein likho)

1. **v2 se accuracy improve nahi hui**, dono modes mein 0.7% kam hui. Parameter changes se gain nahi mila.
2. **Fine-tuned BERT mein overfitting dikh rahi hai:** training loss 0.76 → 0.26 gir raha hai, lekin test accuracy epoch 1 ke baad (72.93%) girti ja rahi hai (71.47%). Model train data yaad kar raha hai.
3. **Frozen BERT jaldi plateau kar jata hai** (~66-67%), matlab sirf [CLS] + linear head utna expressive nahi hai.
4. **Training time lagbhag 2x** badha (frozen 453→868s, fine-tuned 900→2114s), kyunki MAX_LEN 256 aur extra epoch. Accuracy gain ke bina ye cost badhi.
5. Fine-tuning ka gain v2 mein +4.80% hai (66.67 → 71.47), v1 jitna hi.
6. Paper ka text accuracy 91.50% hai, abhi gap ~20% ka hai. Reasons: alag dataset, star-rating-based noisy labels (3-star mixed hote hain), limited tuning.

Ek methodology point (sir pooche toh): epoch-wise accuracy **test set** par nikal rahe ho aur wahi final result bhi hai. Alag validation set nahi hai, toh "best epoch" test set dekhke chunna technically data leakage hoga. Safe hai ki final result last epoch ka hi report karo, ya 70-10-20 train/val/test split karo.

NANTAR MAGH SIR NE SUGGEST KELA METHODS TE APPLY KELA

https://www.kaggle.com/code/aakarsonawane/notebooka68bfcd33a

1. Implementation details

Idea: Ek model (BERT ya RoBERTa) ki jagah dono ko milake fusion model banana, taaki dono ke representations ka faayda mile. Teeno cases frozen aur fine-tuned dono modes mein chalaye gaye.

	Case 3: Concatenation	Case 4: Sequential	Case 5: Cross-Attention
Flow	BERT [CLS] (768) aur RoBERTa [CLS] (768) concat → 1536-d →	BERT ke saare token hidden states → RoBERTa mein inputs_embeds ke roop	RoBERTa ka [CLS] **query**, BERT ke saare token states **key/value** → MultiheadAttention (8

	Case 3: Concatenation	Case 4: Sequential	Case 5: Cross-Attention
	Dropout → Linear(1536→3)	mein → RoBERTa ka position-0 output → Linear(768→3)	heads) → residual + LayerNorm → Linear(768→3)
Tokenizers	2 (BERT + RoBERTa)	1 (sirf BERT)	2
Extra trainable layers	sirf classifier	sirf classifier	cross-attention + LayerNorm + classifier

Common hyperparameters:

Parameter	Frozen	Fine-tuned
MAX_LEN	256	256
Batch size	16 (do backbones memory mein hain)	16
Epochs	4	3
Learning rate	5e-4	2e-5
Scheduler	Linear warmup (10%) + decay	same
Gradient clipping	1.0	1.0
Optimizer / Loss	Adam / CrossEntropy	same
Dropout	0.3	0.3
Seed	42	42

Environment: Kaggle GPU, PyTorch + HuggingFace Transformers, scikit-learn metrics.

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2. Data preprocessing

Same hai pichle notebooks jaisa:

1. Sirf Review + Rating, dropna, strip, empty hataye, duplicates drop
2. Rating → Sentiment: 4-5 Positive, 3 Neutral, 1-2 Negative
3. Balanced sampling: 5000 per class = 15,000 (seed 42)
4. Label encoding: Negative=0, Neutral=1, Positive=2

5. Stratified 80-20 split: 12,000 train / 3,000 test
6. Tokenization, padding/truncation to 256 tokens

Stage	Rows
Original	222,020
After cleaning	199,171
Full distribution	Positive 156,390 / Neutral 22,376 / Negative 20,405
Balanced	15,000
Train / Test	12,000 / 3,000

Naya detail: Case 3 aur 5 mein **dual tokenization** hai (har review ko BERT aur RoBERTa dono ke tokenizer se alag tokenize kiya jata hai). Case 4 mein sirf BERT tokenizer, kyunki RoBERTa ko token ids ki jagah BERT ke hidden states milte hain.

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3. Exploration (EDA)

Same hai: shape, columns, dtypes, missing values, word count, class distribution (full vs balanced). Key finding: dataset **heavily imbalanced** (Positive ~78%), isliye balanced undersampling. Phase 1 ka saved output sirf file listing tak hai, toh word-count aur missing-value numbers is notebook se nahi mil sakte.

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4. Results (test set, 3,000 samples, weighted metrics)

Case	Mode	Accuracy	Precision	Recall	F1	Time (s)
3: Concat	Frozen	67.60	67.31	67.60	67.37	1828.0
3: Concat	Fine-tuned	73.43	73.80	73.43	73.51	3500.6
4: Sequential	Frozen	50.53	50.08	50.53	47.35	1847.4
4: Sequential	Fine-tuned	72.87	72.82	72.87	72.76	3478.5
5: Cross-Attn	Frozen	73.07	73.19	73.07	73.01	1868.8
5: Cross-Attn	Fine-tuned	73.47	73.72	73.47	73.56	3529.7

Epoch-wise (loss / test acc):

Epoch	C3 frozen	C3 fine-tuned	C4 frozen	C4 fine-tuned	C5 frozen	C5 fine-tuned
1	0.933 / 66.00	0.731 / **74.33**	1.095 / 47.43	0.883 / 70.87	0.806 / 70.13	0.780 / 73.60
2	0.798 / 67.00	0.527 / 73.83	1.066 / 49.07	0.658 / 71.60	0.680 / 71.67	0.546 / **74.10**
3	0.779 / 67.40	0.384 / 73.43	1.054 / 51.27	0.546 / **72.87**	0.643 / 72.00	0.400 / 73.47
4	0.772 / 67.60	n/a	1.050 / 50.53	n/a	0.602 / **73.07**	n/a

Comparison with single models (v1 aur v2 notebooks se):

Model	Frozen	Fine-tuned
BERT alone (v1 / v2)	67.37 / 66.67	72.17 / 71.47
RoBERTa alone (v1)	66.37	75.07
Case 3 Concat	67.60	73.43
Case 4 Sequential	50.53	72.87
Case 5 Cross-Attn	**73.07**	73.47

Observations (discussion ke liye)

1. **Best frozen result Cross-Attention ka hai (73.07%)**, jo baaki frozen models se 5.5-6.4% upar hai. Wajah: iske paas trainable cross-attention layer hai, jabki baaki frozen models mein sirf ek linear head train hota hai. Ye frozen backbone ke saath bhi dono models ki jaankari smartly mix kar pata hai.
2. **Sequential frozen fail hua (50.53%).** Reason: BERT ke hidden states RoBERTa ke input embedding space mein nahi baithte (dono ki embedding distribution alag hai), aur RoBERTa frozen hai toh wo adapt nahi kar sakta. Fine-tuning mein ye mismatch theek ho gaya (72.87%), isliye frozen vs fine-tuned ka gap sabse bada (+22.34%) hai.
3. **Fine-tuned mein teeno fusion cases ka accuracy lagbhag barabar hai (72.9-73.5%).** Fusion type ka fine-tuned accuracy par bahut fark nahi pada.
4. **Fusion ne RoBERTa fine-tuned (75.07%, v1) ko beat nahi kiya.** Best fusion fine-tuned (73.47%) usse ~1.6% kam hai. Ye honest finding hai: do models jodne se is

dataset par improvement nahi mila, jabki training time ~4x (3500s vs 906s) ho gaya. Ye comparison v1 ke RoBERTa se hai (MAX_LEN 128, batch 16); fusion notebook mein MAX_LEN 256 hai, toh exact apples-to-apples nahi hai. Report mein ye note zaroor likhna.

5. **Overfitting pattern:** Case 3 fine-tuned mein loss girta rahta hai (0.73 → 0.38), par accuracy epoch 1 ke baad girti hai (74.33 → 73.43). Case 5 fine-tuned bhi epoch 2 ke baad girta hai. Case 4 fine-tuned mein accuracy badhti rahi.
6. **Training time:** fusion models ~2x slower hain single model se (frozen ~1850s, fine-tuned ~3500s), kyunki forward pass mein do backbones chalte hain.
7. **Paper se gap:** paper ka text accuracy 91.50% hai, aapka best 74-75%. Reasons: alag dataset, star-rating based noisy labels, limited tuning.

Methodology note: epoch-wise accuracy test set par nikal rahe ho, aur alag validation set nahi hai. Final result last epoch ka report karo (jo notebook karta hai), best epoch chunoge toh test-set leakage hoga. Ye bhi dhyan rahe ki "test_acc" table mein dikhne wala best-epoch number final result nahi hai.

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5. Visualizations

Har case (3, 4, 5) ke liye 4-4 plots bane hain (inline, files save nahi hui):

Plot	Kya dikhata hai
Loss + Accuracy curves (side by side)	frozen vs fine-tuned, epoch-wise
Accuracy bar chart	frozen vs fine-tuned accuracy
Precision/Recall/F1 grouped bar	teeno metrics mode-wise
Execution time bar	training time mode-wise

Plots **inline** hain, save nahi hue. Blackbook/paper mein daalne ke liye Kaggle se screenshot lo ya plt.savefig() add karke dobara run karo.

Missing (add karne se marks badhenge): teeno cases ka combined comparison chart (ek hi graph mein 6 configs), confusion matrix, per-class metrics, class distribution chart, architecture diagrams.

Paper vs Blackbook

- **Paper:** fusion architectures ka concise description (3 diagrams), hyperparameter table, 6-config results table, single-model vs fusion comparison, discussion points 1, 2, 4, conclusion.
- **Blackbook:** upar ka sab + preprocessing details, har architecture ka forward pass explanation, epoch-wise tables, saare plots, limitations, future scope.

Conclusion (ready to use): *"Among the three BERT-RoBERTa fusion strategies, cross-attention fusion performed best in the frozen setting (73.07%), while all three fusion strategies reached 72.9-73.5% when fine-tuned. However, fusion did not surpass fine-tuned RoBERTa alone (75.07%) on this dataset, and roughly doubled training cost."*

Future scope: validation split ke saath early stopping, hyperparameter tuning, fusion with face-emotion model (multimodal), per-class error analysis.

NANTAR MAGH APAN TEXT MODEL VAR EVALUATION AND TESTING METHOD APPLY KELA TAAKI HAI REAL TIME APPLICATION SAATHI DEPLOYABLE MODEL BANUN ZAILA PAHIJE ACTUALLY

TEXT MODEL EVALUATION AND TESTING - https://www.kaggle.com/code/akruti850/text-model

1. Implementation details

Phase 3: Cross-Attention Fusion model

- BERT: poore token hidden states (B, L, 768)
- RoBERTa: sirf [CLS] (B, 1, 768), jo **query** banta hai
- nn.MultiheadAttention (8 heads): RoBERTa ka CLS, BERT ke saare tokens par attend karta hai (padding masked)
- Residual connection + LayerNorm → Dropout(0.3) → Linear(768→3)
- Total parameters: **236,494,083** (~902 MB fp32)

Hyperparameters:

Parameter	Frozen	Fine-tuned
MAX_LEN	256	256
Batch size	16	16
Epochs	4	3
LR	5e-4	2e-5

Parameter	Frozen	Fine-tuned
Scheduler	Linear warmup 10% + decay	same
Grad clipping	1.0	1.0
Optimizer / Loss	Adam / CrossEntropy	same
Seed	42	42

Phase 4 (4 pillars):

- **A.** Statistical evaluation: classification report, confusion matrix, bootstrap CI, baseline significance, error analysis
- **B.** Real-world probe testing: 12 hand-written sentences (sarcasm, negation, mixed, short text, Manglish, Hinglish, neutral, caps)
- **C.** 9 unit tests (tokenizer, model forward, softmax, determinism, batch consistency, long input)
- **D.** Production readiness: checkpoint save/load, predict_api(), latency benchmark, prediction logging

Phase 5: model ko 6 alag public datasets par test kiya (IMDB, Yelp, Amazon x2, Sentiment140, Twitter Airline) 3,000 random samples har ek se.

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2. Data preprocessing

Main dataset: pehle jaisa hi (Google Reviews, star → sentiment mapping, balanced 5000/class, stratified 80-20, dual tokenization BERT + RoBERTa, max 256 tokens).

Phase 5 ke external datasets ke liye alag preprocessing:

- Text clean (strip, empty hataye, duplicates drop)
- Label mapping: binary datasets (positive/negative, ya 1/2, 0/4) → Negative/Positive; Twitter Airline ke 3 labels seedhe use hue
- 3,000 random samples (seed 42)
- Important: binary datasets mein **Neutral class hai hi nahi**, jabki model 3-class hai

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3. Exploration (EDA) methods

- Phase 1: shape, dtypes, missing values, word count, class distribution. Key finding: **imbalanced data** (Positive 156,390 / Neutral 22,376 / Negative 20,405), isliye balanced sampling.
- Phase 4 mein **error analysis** naya EDA-type tool hai: sabse confident galat predictions nikalna.
- Phase 5 mein har external dataset ka class distribution print hota hai.

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4. Results

Phase 3 (model comparison)

Mode	Accuracy	Precision	Recall	F1	Time (s)
Frozen	73.07	73.19	73.07	73.01	1744.5
Fine-tuned	73.47	73.72	73.47	73.56	3394.7

Epoch-wise (frozen): 70.13 → 71.67 → 72.00 → 73.07. Fine-tuned: 73.60 → **74.10** → 73.47.

Phase 4A: Fine-tuned model ka detailed evaluation

Class	Precision	Recall	F1
Negative	0.81	0.77	0.79
Neutral	0.62	0.64	**0.63**
Positive	0.78	0.80	0.79
Macro avg	0.74	0.73	0.74

- **Bootstrap 95% CI (1000 resamples):** Accuracy 73.44% [71.87 - 74.87], Macro F1 73.53% [71.98 - 74.93]
- **Majority baseline:** 33.33%, model **+40.11 points** upar, CI ke bahar, toh statistically significant
- **Errors:** 796 / 3000 galat (26.5%)

Phase 4B: Probe testing: 11/12 passed (91.7%)

Sirf ek fail: *"Not bad at all, actually quite good."* (expected Positive, predicted Neutral, 80.3% confidence), yaani **negation** wali sentence. Sarcasm, mixed sentiment, Manglish aur Hinglish sab pass hue. Dhyan rahe ye sirf 12 sentences ka chhota hand-made test hai, isse generalization ka strong claim mat karna.

Phase 4C: Unit tests: 9/9 passed

Phase 4D: Production metrics

Metric	Value
Checkpoint size	902.3 MB
Reload prediction matches original	True
Single-request latency	mean 33.7 ms, p50 33.6, p95 34.8, p99 35.3
Batch throughput (16 texts)	30.2 texts/sec

Phase 5: Cross-dataset (out-of-domain) results

Dataset	Accuracy	Macro F1	Weighted F1
Malaysia Restaurant (in-domain)	73.47	73.56	73.56
IMDB Movies	72.67	51.01	76.55
Yelp Polarity	**85.70**	60.37	90.55
Amazon Polarity	79.63	57.42	86.09
Amazon Polarity (yacharki)	79.63	57.42	86.09
Sentiment140 (Twitter)	58.40	43.52	65.28
Twitter US Airline	73.67	62.33	71.30

Observations (discussion ke liye)

1. **Neutral class sabse kamzor hai** (F1 0.63 vs 0.79 baaki dono). 3-star reviews ka sentiment mixed hota hai.
2. **Error analysis ka important finding:** sabse confident galat predictions mein bahut saare reviews “Neutral” label wale hain par text clearly positive ya negative hai (jaise “Delicious! highly recommended!” ya “Fast service!!”, true label Neutral). Ye **label noise** hai: star rating se sentiment map karne ka nuksaan, 3-star reviews ka text hamesha neutral nahi hota. Isliye 73% accuracy ka ek bada hissa model ki galti nahi, label ki limitation hai. Ye report mein strong point banta hai.
3. **Yelp (85.7%) aur Amazon (79.6%) par accuracy in-domain se bhi zyada hai**, kyunki ye binary tasks hain (neutral nahi) aur review style restaurant reviews se

milta-julta hai. Twitter (58.4%) sabse kamzor, kyunki short, slang-heavy tweets alag domain hain.

4. **Binary datasets par Macro F1 misleading hai** (43-60%). Model ko Neutral class ka 0 support milta hai, jisse Neutral ka F1 = 0 aa jata hai aur macro average gir jata hai. In datasets ke liye **accuracy aur weighted F1** dekho, macro F1 nahi. Ye report mein clearly likhna, warna reader galat samjhega.
5. **Positive recall binary datasets mein kam hai** (IMDB 0.54, Yelp 0.83, Amazon 0.75) kyunki kuch positive reviews "Neutral" predict ho jaate hain. Model 3-class hai aur in datasets mein Neutral label hota hi nahi.
6. **Twitter Airline (jisme 3 class hain) sabse fair comparison hai:** 73.67% accuracy, Neutral F1 sirf 0.31 (kamzor), Negative F1 0.85. Wahan class imbalance hai (Negative 1877 / Neutral 647 / Positive 476).
7. **Amazon ke dono rows bilkul same** (79.63 / 57.42 / 86.09). Ye almost certainly **same test file ka duplicate** hai (dono Amazon polarity dataset ke versions), isliye ise ek hi dataset ginna, 6 mein se 5 independent hain.
8. **Latency ~34 ms per request** real-time use ke liye acceptable hai, par 902 MB model size deployment mein heavy hai.
9. **Paper gap:** paper ka text accuracy 91.50%, aapka in-domain 73.47%. Honest reasons: alag dataset, star-rating-based noisy labels, limited tuning.

Methodology notes (sir pooche toh):

- Epoch-wise accuracy test set par hai aur alag validation set nahi hai. Final result last epoch ka (73.47%) hi report karo, fine-tuned ka epoch-2 peak (74.10%) nahi.
- Phase 4 mein "best model" fine-tuned (73.47%) ko chuna gaya, jabki frozen (73.07%) sirf 0.4% peeche hai aur 2x tez train hota hai. Dono ko justify kar sakte ho.
- Confusion matrix ke exact numbers notebook ke text output mein nahi hain (sirf plot hai), toh screenshot se lena padega.

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5. Visualizations

Phase	Plots
Phase 3	Loss + accuracy curves (frozen vs fine-tuned), accuracy bar, precision/recall/F1 grouped bar, execution time bar

Phase	Plots
Phase 4A	**Confusion matrix** (counts + row-normalized heatmap, side by side)
Phase 5	Har external dataset ka confusion matrix (6) + **final cross-dataset comparison bar chart**

Plots inline hain, save nahi hue. Blackbook/paper ke liye screenshot lo ya plt.savefig() add karo.

Abhi bhi missing: class distribution chart, review length histogram, architecture diagram, ek combined chart jisme single models (BERT, RoBERTa) aur teeno fusion cases saath dikhein.

Paper vs Blackbook

- **Paper (concise):** cross-attention architecture diagram, hyperparameter table, Phase 3 results, classification report + confusion matrix, cross-dataset table (Amazon ek hi baar), discussion points 1, 2, 3, 4, conclusion.
- **Blackbook (detailed):** upar ka sab + preprocessing steps, Phase 4 ke saare 4 pillars (probe table, 9 unit tests, production metrics), pehle ke notebooks se single-model aur fusion cases ka comparison, limitations, future scope.

Conclusion (ready to use): *"The BERT-RoBERTa cross-attention fusion model achieved 73.47% accuracy (95% CI: 71.87-74.87%) on 3-class restaurant review sentiment, 40 points above the majority baseline. It generalized well to Yelp (85.7%) and Amazon (79.6%) polarity data, passed 11/12 hand-crafted probe cases and 9/9 unit tests, with ~34 ms inference latency. The Neutral class remained the weakest (F1 0.63), largely due to star-rating label noise."*

Future scope: better neutral labeling (sentiment-score thresholds like the paper instead of star ratings), validation split with early stopping, model distillation to cut the 902 MB size, aur face-emotion model ke saath multimodal fusion.

```
======================================================================
FINAL COMPARISON ACROSS ALL DATASETS
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Dataset	N	Accuracy %	Macro F1 %	Weighted F1 %
Malaysia Restaurant Reviews (in-domain)	3000	73.47	73.56	73.56
IMDB Movie Reviews (50K)	3000	72.67	51.01	76.55
Yelp Review Polarity	3000	85.70	60.37	90.55
Amazon Reviews Polarity	3000	79.63	57.42	86.09
Amazon Reviews Polarity (yacharki)	3000	79.63	57.42	86.09
Sentiment140 (Twitter)	3000	58.40	43.52	65.28
Twitter US Airline Sentiment	3000	73.67	62.33	71.30

